

SYNOPSIS 1

E-Commerce Customer Purchase Analysis Using Power BI

1. Introduction

E-commerce platforms generate a large volume of transactional data related to customer purchases, product categories, and geographic markets. Analyzing this data is essential for understanding customer behavior and improving business strategies.

This project focuses on analyzing customer purchasing behavior using Power BI. By visualizing sales data across countries, product categories, and customer demographics, the dashboard provides valuable insights into revenue generation and customer spending patterns.

2. Business Problem

A global retail company wants to analyze customer purchasing behavior across different countries, genders, and product categories. The organization aims to understand which product categories generate the highest revenue, which countries contribute the most sales, and which customer segments spend the most.

Additionally, the company wants to analyze how sales trends change over time in order to identify patterns in customer purchasing behavior. The main objective is to identify high-value customers, profitable markets, and top-performing product categories that can help improve marketing strategies and drive revenue growth.

3. Objectives of the Project

The main objectives of this project include:

- To analyze customer purchasing patterns across different countries.
- To identify the most profitable product categories.
- To analyze spending behavior across different customer demographics such as age and gender.
- To track sales trends over time.
- To develop an interactive Power BI dashboard that provides meaningful business insights.

4. Key Performance Indicators (KPIs)

The dashboard includes the following key performance indicators:

Total Revenue

Represents the total revenue generated from all purchases.

Total Customers

Shows the total number of unique customers.

Average Purchase Value

Represents the average revenue generated per customer.

Total Transactions

Indicates the total number of purchase records.

5. Chart Requirements

The dashboard includes several visualizations to analyze sales performance from different perspectives.

Revenue by Product Category (Bar Chart)

This chart identifies which product categories generate the highest revenue.

Revenue by Country (Map Visualization)

This visualization highlights the countries contributing the most to overall sales.

Gender Distribution (Donut Chart)

This chart shows the distribution of customers based on gender.

Sales Trend Over Time (Line Chart)

This visualization tracks changes in sales over time to identify trends and growth patterns.

Age vs Spending (Scatter Chart)

This chart analyzes how spending behavior varies across different age groups.

Country vs Category Sales (Stacked Bar Chart)

This visualization compares product category performance across different countries.

6. Expected Outcomes

The dashboard provides insights that help businesses:

- Identify high-value customer segments.
- Determine top-performing product categories.
- Analyze market performance across different countries.
- Understand demographic spending patterns.
- Monitor sales trends and revenue growth over time.

These insights support better decision-making for marketing strategies and product offerings.

7. Tools and Technologies Used

- **Power BI Desktop** – Data visualization and dashboard development
- **Microsoft Excel** – Dataset storage and preprocessing
- **DAX (Data Analysis Expressions)** – Calculation of key performance indicators

8. Output



9. Conclusion

This project demonstrates how data analytics can help businesses understand customer behavior and market trends. Using Power BI, the e-commerce sales data was transformed into an interactive dashboard that highlights key insights related to revenue, demographics, and product performance. These insights enable organizations to make data-driven decisions to improve sales and customer engagement.

SYNOPSIS 2

Supply Chain Inventory Analysis Using Power BI

1. Introduction

Efficient supply chain management is essential for ensuring product availability and maintaining smooth business operations. Companies often manage multiple warehouses, suppliers, and products, making inventory monitoring a complex task. Data analytics plays a critical role in analyzing inventory levels, supplier performance, and product demand.

This project focuses on analyzing supply chain inventory data using Power BI. The dashboard provides insights into warehouse stock levels, supplier performance, and product demand patterns to support effective supply chain management.

2. Business Problem

A retail company manages multiple warehouses and suppliers. The company needs to monitor inventory levels, supplier efficiency, and product demand to ensure smooth supply chain operations.

Management requires insights to determine which warehouse holds the most stock, which supplier delivers products the fastest, and which products have the highest demand. Additionally, the company needs to evaluate whether current stock levels are sufficient to meet customer orders.

The objective is to optimize inventory distribution, reduce delivery delays, and ensure product availability across warehouses.

3. Objectives of the Project

The main objectives of this project are:

- To analyze inventory distribution across different warehouses.
- To evaluate supplier performance based on delivery time and order contribution.
- To identify products with the highest demand.
- To compare available stock with order quantities.
- To create a Power BI dashboard that supports supply chain decision-making.

4. Key Performance Indicators (KPIs)

The dashboard includes several key metrics to monitor supply chain performance.

Total Stock Quantity

Represents the total available inventory across all warehouses.

Total Order Quantity

Indicates the total demand from customers.

Stock vs Order Ratio

Shows whether available inventory is sufficient to meet demand.

Average Stock per Product

Represents the average stock quantity maintained for each product.

Average Delivery Time

Measures supplier efficiency based on delivery performance.

5. Chart Requirements

The dashboard includes the following visualizations:

Stock Distribution by Warehouse (Clustered Column Chart)

Shows which warehouse holds the highest inventory.

Supplier Contribution (Donut Chart)

Displays the share of orders supplied by each supplier.

Delivery Time by Supplier (Bar Chart)

Compares supplier performance based on delivery time.

Stock vs Orders Comparison (Clustered Column Chart)

Analyzes whether current stock levels meet customer demand.

Warehouse vs Orders (Treemap)

Shows how order quantities are distributed across warehouses.

Product Demand Analysis (Table/Matrix)

Provides a detailed comparison of product demand, stock levels, delivery time, and suppliers.

6. Expected Outcomes

The dashboard provides insights that help management:

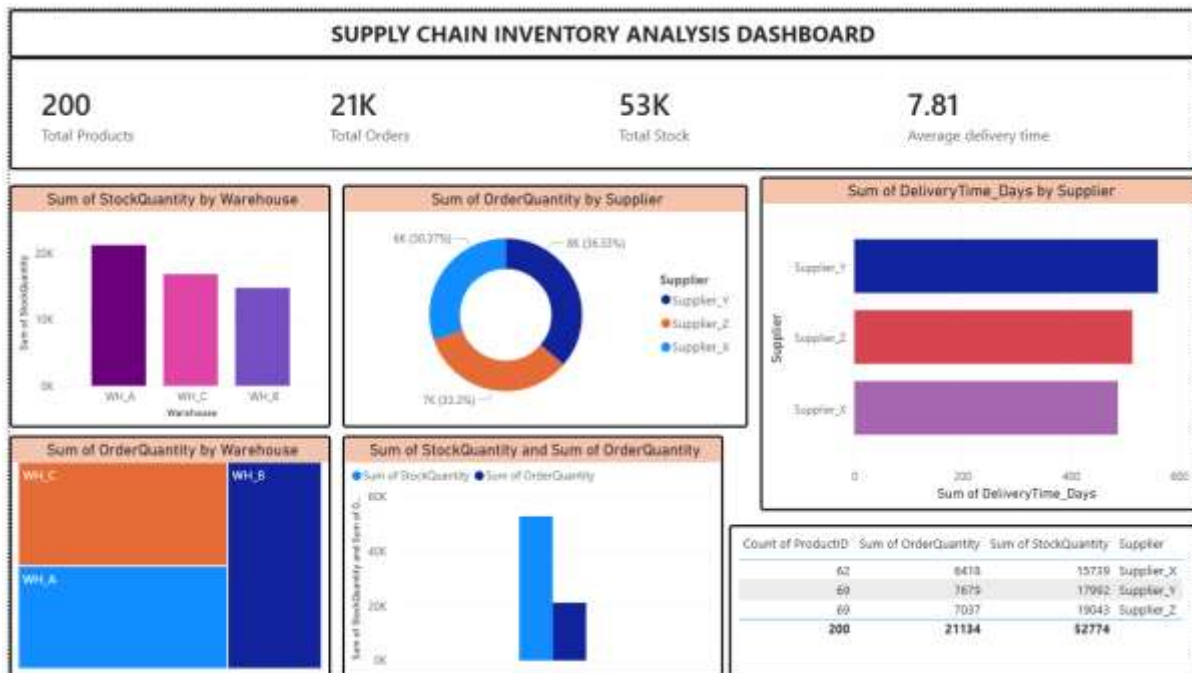
- Identify warehouses with the highest inventory levels.
- Evaluate supplier performance and delivery efficiency.
- Detect products with high demand.
- Identify potential stock shortages.
- Improve inventory planning and distribution.

These insights help organizations optimize supply chain operations and reduce operational inefficiencies.

7. Tools and Technologies Used

- **Power BI Desktop** – Data visualization and dashboard development
- **Microsoft Excel** – Dataset storage and preprocessing
- **DAX (Data Analysis Expressions)** – Calculation of KPIs
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8. Output:



9. Conclusion

This project highlights the importance of data analytics in supply chain management. By transforming raw inventory data into meaningful visual insights using Power BI, the dashboard enables businesses to monitor stock levels, evaluate supplier performance, and optimize inventory distribution. The insights gained from this analysis support better operational planning and improved supply chain efficiency.